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CO2-AT1 – Analytical Problem solving

1. Perception model – Pass or Fail

Given that

Study hours (x_1) = 2

Attendance (x_2) = 1

weights : $w_1 = 0.6$, $w_2 = 0.4$

Bias = -0.5

step-1 : calculate net input

Formula :

$$Net = (x_1 w_1) + (x_2 w_2) + b$$

Substitute;

$$Net = (2 \times 0.6) + (1 \times 0.4) + (-0.5)$$

$$= 1.2 + 0.4 - 0.5$$

$$\boxed{Net = 1.1}$$

step-2 : Apply the step activation function.

* If net input is greater than or eaual to 0, output = 1
* If net input is less than 0, output = 0

since $1.1 \geq 0$ the output is 1.

FINAL ANSWER: The student Passes, because the perceptior output is 1.

2. Backpropagation - weight update

Given that:

Initial weight = 0.5

Learning rate (η) = 0.1

Input = 2

Actual output = 0.6

Target output = 1

Step-1: calculate the Error

Formula:

$$\text{Error} = \text{Target} - \text{Actual}$$

$$\text{Error} = 1 - 0.6$$

$$= 0.4$$

Step-2: calculate the New weight update

Formula:

$$\Delta w = \eta \times \text{Error} \times \text{Input}$$

$$\Delta w = 0.1 \times 0.4 \times 2$$

$$\Delta w = 0.08$$

Step-3 :- calculate the New weight

$$\text{New weight} = \text{old weight} + \Delta w$$

$$= 0.5 + 0.08$$

$$= 0.58$$

FINAL ANSWER: The error is 0.4, the weight update is 0.08, and the new weight is 0.58

3. Confusion Matrix - Model Evaluation

Given that:

True Positive (TP) = 180

True Negitive (TN) = 150

False Positive (FP) = 20

False Negitive (FN) = 50

Total:

$180+150+20+50 = 400$

step-1: calculate accuracy

$$\text{Accuracy} = \frac{TP+TN}{\text{Total}} \times 100$$

$$= \frac{180+150}{400} \times 100$$

$$= 82.5\%$$

step-2: calculate precision

$$\text{precision} = \frac{TP}{TP+FP} \times 100$$

$$= \frac{180}{180+20} \times 100$$

$$= 90\%$$

step-3 : calculate sensitivity / recall

$$\text{Recall} = \frac{TP}{TP+FN} \times 100$$

$$= \frac{180}{180+50} \times 100$$

$$= 78.26\%$$

step-4 : calculate specificity

$$\text{specificity} = \frac{TN}{TN+FP} \times 100$$

$$= \frac{150}{150+20} \times 100$$

$$= 88.24\%$$

4A Genetic Algorithm - Fitness and crossover

Given that :

Function :

$$f(x) = x^2$$

Initial Population : 2, 4, 6, 8

step-1 : calculate fitness values

Individual	$f(x)=x^2$
2	4
4	16
6	36
8	64

Step-2: select the best candidates

- First candidate: 8, fitness = 64
- second candidate: 6, fitness = 36

Step-3: Perform crossover

Conver to binary

* 8 = 1000
* 6 = 0110

Exchange their significant

→ off spring 1 = 1000 = 8
→ off spring 2 = 0110 = 6

Step-4: Explain mutation

Importance :-

i) Maintains diversity in the Population
ii) Helps Explore new solutions
iii) Reduces the chance of getting stuck at a local optimum

5. Deep Neural Network overfitting

Given

- Training accuracy increases from 60% to 95%
- validation accuracy increases from 58% to 72%. then decreases to 68%

Step-1: Identify the Problem

→ The problem is overfitting

Step-2: Explain why it occurs

→ overfitting happens when a model learns the training data too closely, including noise and unnecessary details.

Step-3: suggest two techniques

1. Dropout: Randomly disables some neurons during training to reduce overdependence on particular neurons

2. Early stopping: stops training when validation performance begins to worsen, helping prevent overfitting

Step-4: Explain validation accuracy

→ validation accuracy measures how well the model performs on data that is not used to train it. it helps identify overfitting and shows whether the model can generalize to new data.